

LinearSkills Academy

Class 7 — while Loop
Zero to Advance

Part 1 — The Basics

What is a while loop?

A loop that keeps running as long as a condition is true. Used when you don't know in advance how many times it should run.

1. Basic while Loop

```
i = 1
while i <= 5:
    print(i)
    i = i + 1
```

Output: 1 2 3 4 5. The counter (i) must be updated manually, or the loop will never stop.

2. break — stop the loop

```
i = 1
while i <= 10:
    if i == 5:
        break
    print(i)
    i = i + 1
```

Output: 1 2 3 4. As soon as `i == 5`, the loop stops immediately.

3. continue — skip this round

```
i = 0
while i < 5:
    i = i + 1
    if i == 3:
        continue
    print(i)
```

Output: 1 2 4 5. When `i == 3`, the print is skipped but the loop keeps going.

4. Infinite Loop

```
while True:
    print("This runs forever")
```

If the condition never becomes False, the loop will never stop. Use `break` to exit it:

```
while True:
    answer = input("Type stop to exit: ")
    if answer == "stop":
        break
```

5. for vs while — Quick Difference

Feature	for loop	while loop
Use case	You know how many times	Depends on a condition
Counter	Automatic (range)	Must be updated manually
Infinite loop risk	Low	High (if done wrong)

Part 2 — while ... else (Intermediate to Advance)

6. The Basic Idea

A while loop can have an else block attached to it. Rule: else runs only if the loop finished normally (condition became False on its own). It does NOT run if the loop was stopped early using break.

```
while condition:
    # loop body
else:
    # else body
```

7. Example — No break

```
i = 1
while i <= 3:
    print(i)
    i += 1
else:
    print("Loop finished completely")
```

Output: 1, 2, 3, then "Loop finished completely" — the loop ran fully, so else executed.

8. Example — With break

```
i = 1
while i <= 3:
    if i == 2:
        break
    print(i)
    i += 1
else:
    print("Loop finished completely")
```

Output: only 1. The else message is NOT printed, because break cut the loop short.

9. Why does this exist? (Real use case)

while-else is most useful for search-style logic — checking if something was found or not, without needing an extra flag variable.